

V Madhan Kumar

Bengaluru, Karnataka, India

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 GitHub  Portfolio

Education

National Institute of Technology, Karnataka, Surathkal

BTech in Computer Science Engineering

Minors in Management

2019 – 2023

CGPA: 8.47

CGPA: 8.6

Experience

Jumbo (JoinJumbo), Delhi — *Backend Developer*

Sep 2025 – Present

- **Scalable Backend APIs & Microservices** — Designed and built backend services across 10+ microservices handling payments, fantasy sports, and user management using Node.js/TypeScript on AWS Lambda, serving high-traffic transactional workloads with 1,200+ commits.
- **Redis Caching & Data Infrastructure** — Architected a Redis-first caching layer (AWS ElastiCache) with database fallback for high-read APIs, implementing granular cache invalidation, cache warming, and CROSSSLOT error resolution — reducing p99 latency from 128ms to sub-30ms and eliminating 70–90% of database queries on hot paths.
- **ClickHouse Integration for Analytics** — Integrated ClickHouse as an analytics backend for batch point/rank computation, enabling efficient large-scale data processing alongside PostgreSQL.
- **SQL & Database Optimization** — Managed Aurora PostgreSQL clusters, tuned parameter groups, added strategic indexes, optimized query patterns (EXISTS vs FindOne, query builder refactors), and monitored performance via CloudWatch Performance Insights.
- **Observability & Monitoring** — Designed and implemented Datadog APM integration (dd-trace) with distributed tracing, log correlation, and error tracking across multiple services; operated dual instrumentation alongside SigNoz/OpenTelemetry for validation before production cutover.
- **Data Pipeline & Payment Architecture** — Built end-to-end payment pipelines (Razorpay + Slice) with SQS async processing, webhook ingestion, idempotency checks, gateway switching for A/B testing, atomic wallet reversals, and event-driven state machines via EventBridge.

PlayGPL (Litifer), Bengaluru — *Software Developer*

May 2023 – Mar 2025

- **Backend Systems & API Development** — Designed scalable REST APIs and microservices using Golang and Node.js, implementing authentication, authorization, rate limiting, and analytics for high-traffic transactional systems.
- **Fraud Detection Data Pipeline** — Built fraud detection system using device fingerprints, network behavior analysis, and root/VPN checks — processing user signals to prevent referral abuse and malicious transactions.
- **Full-Stack Development** — Developed mobile apps (Flutter) with deep linking, push notifications, and user tracking; optimized web applications with Next.js and server-side rendering.

Hoggy (Litifer), Bengaluru — *Flutter Developer Intern*

May 2022 – July 2022

- Built and maintained Android & iOS apps using Flutter. Integrated REST APIs and conducted end-to-end testing.

Projects

SQL Query Analyzer & Optimizer

 GitHub

FastAPI, sqlglot, Claude API, PostgreSQL, Docker

- Built a FastAPI service that parses SQL queries using sqlglot to extract tables, columns, joins, and subqueries, then detects performance anti-patterns (SELECT *, missing WHERE, excessive joins) and computes a complexity score.
- Integrated Claude API for AI-powered optimization suggestions (index definitions, query rewrites); includes rule-based fallback, data lineage tracking with graph representation, and paginated query history.

Technical Skills

- **Languages:** Python, Golang, Node.js/TypeScript, JavaScript, SQL, Dart, C++
- **Frameworks:** FastAPI, Express, NestJS, Flutter, React, Next.js
- **Cloud & Infra:** AWS (Lambda, Aurora/RDS, SQS, EventBridge, ElastiCache, S3, API Gateway, IAM), Kubernetes (EKS/DOKS), Docker, Cloudflare WAF
- **Databases:** PostgreSQL (Aurora), Redis, ClickHouse, SQLite
- **AI/ML:** AWS Rekognition, LangChain/RAG (exploring), LLM-assisted development workflows
- **Tools:** Git, TypeORM, Datadog APM, SigNoz/OpenTelemetry, CloudWatch, Grafana, GitHub Actions CI/CD